



Founded 1982

The Hills Grammar School

2013 Higher School Certificate Trial Examination

Mathematics

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen.
Black pen is preferred.
- Board-approved calculators may be used.
- A table of standard integrals is provided at the back of this paper which may be detached and used throughout the paper.
- In Questions 11-16, show relevant mathematical reasoning and/or working.

Total marks – 100

Section I

10 marks

- Attempt Questions 1-10
- Answer on the Multiple Choice Answer Sheet provided.
- Allow about 15 minutes for this Section.

Section II

90 marks

- Attempt Questions 11-16
- Answer in the writing booklets provided.
- Start a new booklet for each question.
- Allow about 2 hours and 45 minutes for this Section.

Section I
Total marks (10)
Attempt Questions 1-10
Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet provided.
Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Example: $2 + 4 = ?$

- (A) 2
- (B) 6
- (C) 8
- (D) 9

A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word *correct* and drawing an arrow as follows:

A ☒ B ☒ C ☐ D ☐

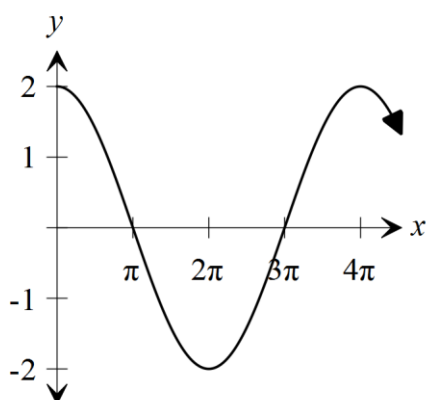
1 The gradient of the line $2x - 3y + 7 = 0$ is:

- (A) 2
- (B) $\frac{2}{3}$
- (C) $-\frac{2}{3}$
- (D) $\frac{3}{2}$

2 The conditions for the expression $ax^2 + bx + c = 0$ to be positive definite are:

- (A) $a > 0$ and $\Delta > 0$
- (B) $c > 0$ and $\Delta > 0$
- (C) $c > 0$ and $\Delta < 0$
- (D) $a > 0$ and $\Delta < 0$

3



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Which equation would best describe the trigonometric graph shown above?

- (A) $y = \frac{1}{2} \sin 2x$
- (B) $y = 2 \sin 2x$
- (C) $y = 2 \cos 2x$
- (D) $y = 2 \cos \frac{x}{2}$

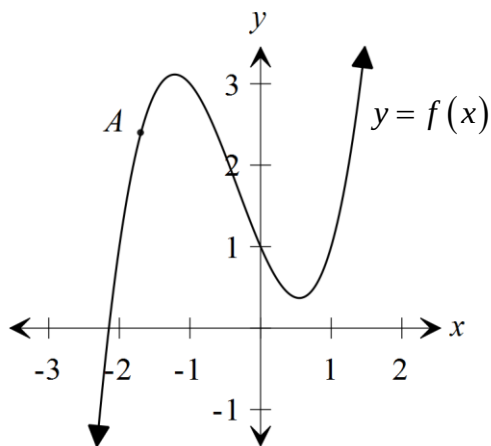
4 What is the domain and range of the function $y = \sqrt{9 - x^2}$?

- (A) Domain $-3 \leq x \leq 3$ Range $0 \leq y \leq 3$
(B) Domain $-3 \leq x \leq 3$ Range $-3 \leq y \leq 3$
(C) Domain $0 \leq x \leq 3$ Range $-3 \leq y \leq 3$
(D) Domain $0 \leq x \leq 3$ Range $0 \leq y \leq 3$

5 What is the value of $\lim_{x \rightarrow 2} \frac{x^2 - 4}{x - 2}$?

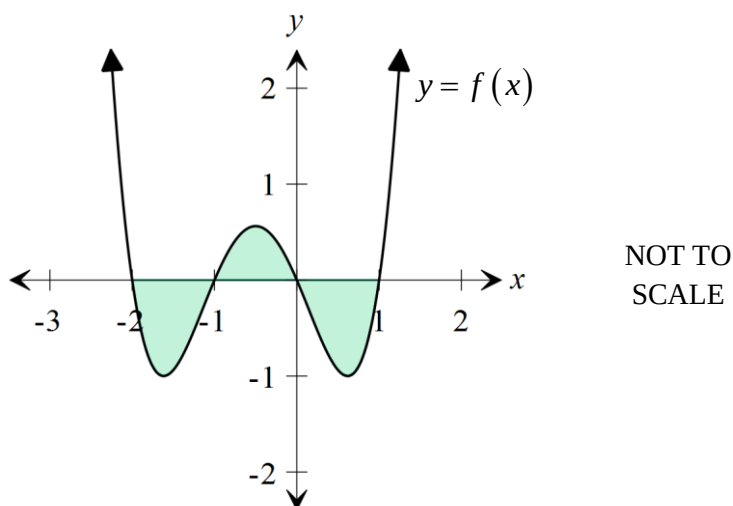
- (A) 0
(B) undefined
(C) 4
(D) 6

6 What properties exist at the point A on the graph of $y = f(x)$ shown below?



- (A) $f'(x) > 0$ and $f''(x) > 0$
(B) $f'(x) > 0$ and $f''(x) < 0$
(C) $f'(x) < 0$ and $f''(x) > 0$
(D) $f'(x) < 0$ and $f''(x) < 0$

- 7 The diagram below shows the graph of $y = f(x)$. Which formula would give the best approximation of the shaded area between $y = f(x)$ and the x axis?



- (A) $A = \int_{-2}^1 f(x) dx$
- (B) $A = \left| \int_{-2}^0 f(x) dx \right| + \int_0^1 f(x) dx$
- (C) $A = 2 \int_{-2}^{-1} f(x) dx + \int_{-1}^0 f(x) dx$
- (D) $A = \int_{-1}^0 f(x) dx + 2 \left| \int_{-2}^{-1} f(x) dx \right|$

- 8 What is the value of $\int_1^{\sqrt{3}} \frac{2dx}{x}$ in simplest form?

- (A) $\ln 2$
- (B) $2\ln\sqrt{3} - 2\ln 1$
- (C) $(\ln\sqrt{3})^2$
- (D) $\ln 3$

- 9 Ten milligrams of caffeine is placed in hot water and begins to dissolve. After t minutes the amount A mg of undissolved caffeine is given by $A = 10e^{-kt}$. What is the value of k given that $A = 3.6$ and $t = 5$?
- (A) -0.717
- (B) -0.204
- (C) 0.204
- (D) 0.717
- 10 In the arithmetic sequence $1, 3 + x, \dots$ what will be the 20th term?
- (A) $(3 + x)^{20}$
- (B) $(3 + x)^{19}$
- (C) $19x + 39$
- (D) $20x + 41$

End of Section I

Section II

Total marks (90)

Attempt Questions 11-16

Start a new booklet for each Question

Allow about 2 hours and 45 minutes for this section

START A NEW BOOKLET

Question 11 (15 marks)

Marks

-
- (a) Evaluate $\sqrt{\frac{\pi}{0.12 \times 3.16}}$ correct to 2 significant figures. 2
- (b) Express 225° in radians in terms of π . 1
- (c) Rationalise the denominator of $\frac{2}{\sqrt{3}+1}$. Give your answer in simplest form. 2
- (d) Simplify $\frac{3x-9}{x^2-9}$. 2
- (e) Find the exact values of θ such that $\tan \theta = \frac{-1}{\sqrt{3}}$, where $0 \leq \theta \leq 2\pi$. 2
- (f) Solve $9^x = \frac{1}{27}$. 2
- (g) Find the coordinates of the focus of the parabola $x^2 = 8(y+1)$. 2
- (h) Solve $|2x-1| > 8$. 2

End of Question 11

START A NEW BOOKLET
Question 12 (15 marks)

Marks

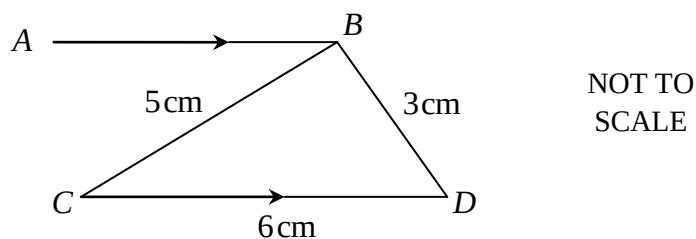
(a) Differentiate with respect to x

(i) $\sqrt{x}$ 1

(ii) $\tan(e^{2x})$ 2

(iii) 4^x . 2

(b)



In the diagram, AB is parallel to CD , CB is 5 cm, CD is 6 cm, BD is 3 cm.

(i) Use the cosine rule to find the size of $\angle BDC$, to the nearest degree. 2

(ii) Hence find the size of $\angle ABD$, giving reasons. 2

(c) Solve $\log_2(5x+1) = 4$. 2

(d) Find:

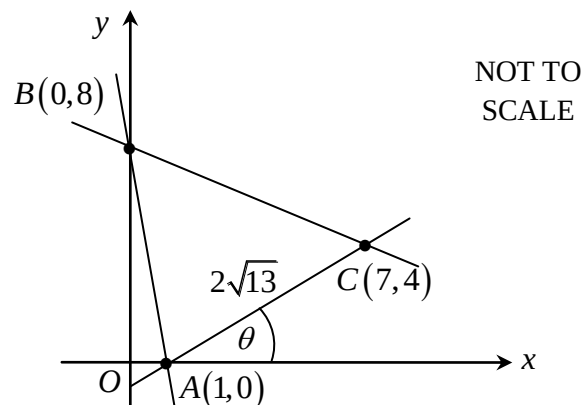
(i) $\int (2x-1)^3 dx$ 2

(ii) $\int_0^{\ln 2} e^{-x} dx$. 2

End of Question 12

START A NEW BOOKLET**Question 13 (15 marks)****Marks**

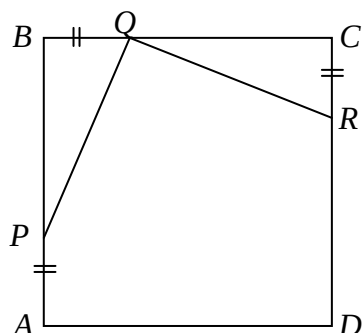
- (a) The points A , B and C have coordinates $(1,0)$, $(0,8)$ and $(7,4)$, as shown in the diagram below. The angle between the line AC and the x axis is θ . $AC = 2\sqrt{13}$.



- | | |
|--|---|
| (i) Find the gradient of the line AC . | 1 |
| (ii) Calculate the size of angle θ to the nearest degree. | 1 |
| (iii) Find the equation of the line AC . | 1 |
| (iv) Find the coordinates of D , the midpoint of AC . | 1 |
| (v) Show that AC is perpendicular to BD . | 1 |
| (vi) Find the area of $\triangle ABC$. | 2 |
| (vii) Write down the coordinates of a point E such that $ABCE$ is a rhombus. | 1 |

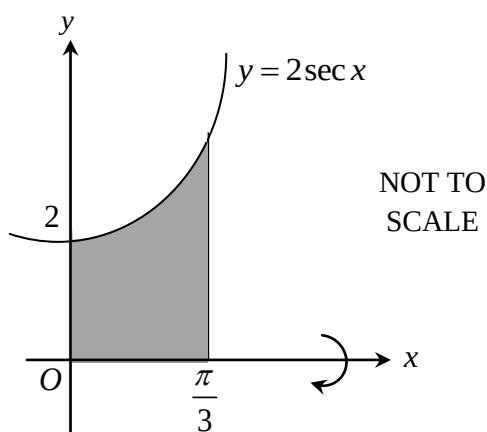
Question 13 continues on the next page

- (b) In the diagram $ABCD$ is a square. The points P , Q and R lie on AB , BC and CD respectively such that $AP = BQ = CR$.



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- (i) Prove that $\triangle PBQ$ and $\triangle CQR$ are congruent. 2
- (ii) Prove that $\angle PQR$ is a right angle. 2
- (c) In the diagram the shaded region is bounded by the curve $y = 2 \sec x$, the coordinate axes and the line $x = \frac{\pi}{3}$. The shaded region is rotated about the x -axis.



Calculate the exact volume of the solid of revolution formed. 3

End of Question 13

START A NEW BOOKLET

Question 14 (15 marks)

Marks

(a) If α and β are the roots of $2x^2 - 3x - 4 = 0$, find the value of

(i) $\alpha + \beta$ **1**

(ii) $\alpha\beta$ **1**

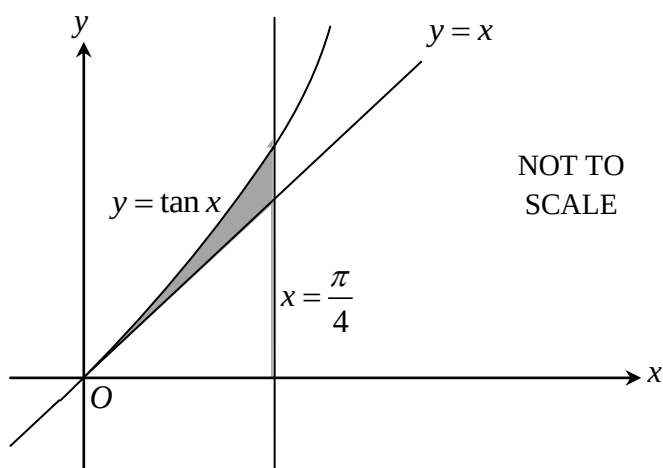
(iii) $\frac{1}{\alpha} + \frac{1}{\beta}$ **1**

(iv) $\alpha^2 + \beta^2$. **2**

(b) Prove that $\sec^2 x + \sec x \tan x = \frac{1}{1 - \sin x}$ **2**

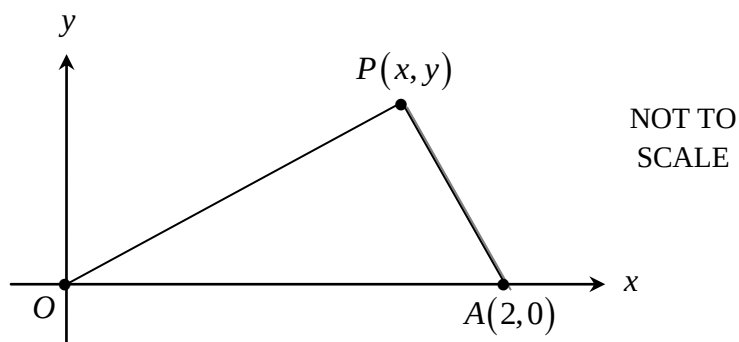
(c) (i) Show that $\frac{d}{dx} \log_e (\cos x) = -\tan x$. **1**

(ii) The shaded region in the diagram is bounded by the curve $y = \tan x$ and the lines $y = x$ and $x = \frac{\pi}{4}$. Using the result from part (i), or otherwise, find the area of the shaded region. **3**



Question 14 continues on the next page

(d)



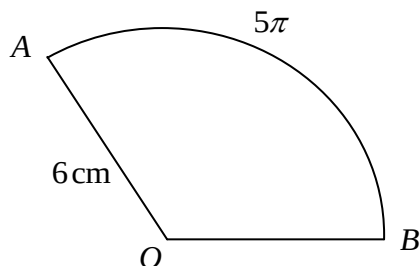
- (i) Show that the equation of the locus of all points P such that OP is perpendicular to AP is $x^2 - 2x + y^2 = 0$.
- (ii) Hence determine the centre and radius of the locus in part (i).

2

2

End of Question 14

(a)



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AOB is a sector of a circle, centre O and radius 6 cm . The length of the arc AB is 5π .
Calculate the area of sector AOB .

2

- (b) A ball is dropped from a height of 4 metres onto a hard floor and bounces. After each bounce, the maximum height reached by the ball is 75% of the previous maximum height. Thus after it first hits the floor, it reaches a height of 3 metres before falling again, and after the second bounce it reaches a height of 2.25 metres before falling again.

- (i) What kind of sequence is formed by successive maximum heights?

1

- (ii) What is the total distance travelled by the ball from the time it was first dropped until it eventually comes to rest on the floor?

2

- (c) (i) Sketch the graph of $y = \log_e x$ and shade the region defined by $\int_1^5 \log_e x \, dx$

2

- (ii) Use Simpson's rule with three function values to estimate $\int_1^5 \log_e x \, dx$

2

Give your answer correct to 2 decimal places.

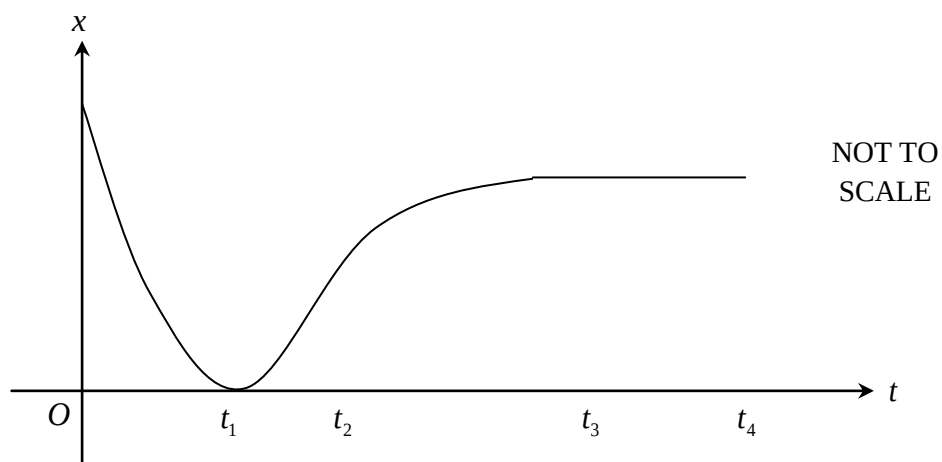
- (d) The curve $y = ax^3 - 9x^2 + b$ has a minimum turning point at $(3, -12)$.

Find the values of a and b .

3

Question 15 continues on the next page

- (e) A particle moves in a straight line and the graph shows the distance x of the particle from a fixed point at time t .



- (i) What is the velocity at $t = t_3$? 1
- (ii) Sketch the graph of velocity v as a function of time. 2

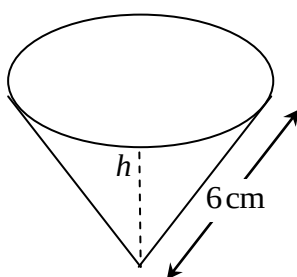
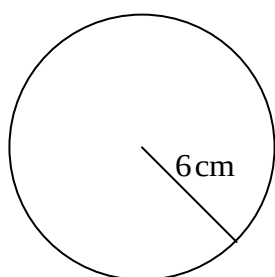
End of Question 15

START A NEW BOOKLET

Question 16 (15 marks)

Marks

- (a) Selina invests \$400 in a bank account at the beginning of each month for 8 years. Interest is to be paid at a rate of 6% per annum compounded monthly.
- (i) Find the amount in the account at the end of the first month. 1
- (ii) Show that the total value of her investment at the end of n years is given by $\$400(1.005 + 1.005^2 + 1.005^3 + \dots + 1.005^{12n})$ 1
- (iii) Find the final value of Selina's investment at the end of the 8 years. Give your answer correct to the nearest dollar. 3
- (iv) What **single** investment at the beginning of the 8 years, with interest compounded monthly, would achieve the same final value? Answer to the nearest dollar. 2
- (b) A circular filter paper of radius 6 cm is cut once along the radius and then each edge formed by this cut is overlapped to make a conical filter with height h and radius r .



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- (i) Show that the volume, V , of the cone is $\frac{1}{3}\pi r^2\sqrt{36-r^2}$ where r is the base radius. 2
- (ii) Show that $\frac{dV}{dr} = \frac{2\pi r\sqrt{36-r^2}}{3} - \frac{\pi r^3}{3\sqrt{36-r^2}}$ 3
- (iii) Find the maximum volume of the cone and the corresponding radius in exact form. 3

End of Examination Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x$, $x > 0$

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Student Number

Multiple Choice Answer Sheet

Section I

Total marks (10)

Attempt Questions 1-10

Allow about 15 minutes for this section

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

- | | | | | |
|----|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | ☐ A | ☐ B | ☐ C | ☐ D |
| 2 | ☐ A | ☐ B | ☐ C | ☐ D |
| 3 | ☐ A | ☐ B | ☐ C | ☐ D |
| 4 | ☐ A | ☐ B | ☐ C | ☐ D |
| 5 | ☐ A | ☐ B | ☐ C | ☐ D |
| 6 | ☐ A | ☐ B | ☐ C | ☐ D |
| 7 | ☐ A | ☐ B | ☐ C | ☐ D |
| 8 | ☐ A | ☐ B | ☐ C | ☐ D |
| 9 | ☐ A | ☐ B | ☐ C | ☐ D |
| 10 | ☐ A | ☐ B | ☐ C | ☐ D |

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Student Number



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The Hills Grammar School

2013 Higher School Certificate Trial Examination

Mathematics

Question	Algebra and Number	Differential Calculus	Functions	Geometry	Integral Calculus	Trigonometry	Total
1-10	10 /1	5, 6, 9 /3	1, 2, 4 /3		7, 8 /2	3 /1	/10
11	(a), (c), (d), (f), (h) /10		(g) /2			(b), (e) /3	/15
12	(c) /2	(a) /5		(b)(ii) /2	(d) /4	(b)(i) /2	/15
13			(a) /8	(b) /4	(c) /3		/15
14		(c)(i) /1	(a), (d) /9		(c)(ii) /3	(b) /2	/15
15	(b) /3	(d), (e) /6			(c) /4	(a) /2	/15
16	(a) /7	(b) /8					/15
Total	/23	/23	/22	/6	/16	/10	/100

THGS Mathematics
THSC 2013 Solutions

Question 11

(a) 2.8783... ✓
2.9 (2 sig figs) ✓
(b) $225 \times \frac{\pi}{180} = \frac{5\pi}{4}$ ✓
(c) $\frac{2}{\sqrt{3}+1} \times \frac{\sqrt{3}-1}{\sqrt{3}-1} = \frac{2\sqrt{3}-2}{2}$
 $= \sqrt{3}-1$ ✓

(d) $\frac{3(x-3)}{(x-3)(x+3)} = \frac{3}{x+3}$ ✓

(e) 
aux $L = \frac{\pi}{6}$ ✓
 $\theta = \pi - \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$
 $\theta = \frac{5\pi}{6}, \frac{11\pi}{6}$ ✓

(f) $3^{2x} = 3^{-3}$ ✓
 $\therefore 2x = -3$
 $x = -\frac{3}{2}$ ✓

(g) $x^2 = 8(y+1)$
 $4a = 8$
 $a = 2$ ✓
focus: $(0, 1)$ ✓

(h) $|2x-1| > 8$
 $2x-1 > 8 \quad 2x-1 < -8$
 $2x > 9 \quad 2x < -7$
 $x > \frac{9}{2} \quad \text{and} \quad x < -\frac{7}{2}$
✓ ✓

Multiple Choice

1. B 2. D 3. D 4. A 5. C 6. B
7. D 8. D 9. C 10. C

Question 12

(a)(i) $\frac{d}{dx}(x^{\frac{1}{2}}) = \frac{1}{2}x^{-\frac{1}{2}}$ or $\frac{1}{2\sqrt{x}}$ ✓

(ii) $\frac{d}{dx}(\tan e^{2x}) = 2e^{2x} \sec^2 e^{2x}$ ✓

(iii) $\frac{d}{dx}(4^x) = \frac{d}{dx}(e^{\ln 4^x}) = \ln 4 e^{\ln 4^x}$
 $= \ln 4 \cdot 4^x$ ✓

(b)(i) let $\angle BDC = \theta$

$\cos \theta = \frac{6^2 + 3^2 - 5^2}{2(6)(3)}$ ✓

$\theta = 56.25^\circ$ or $56^\circ 15'$

$\theta = 56^\circ$ ✓

(ii) $\angle ABD = 180 - 56$ (cont. L_s ✓
(add to 180° ; $AB \parallel CD$)
 $= 124^\circ$ ✓

(c) $2^4 = 5x+1$ ✓

$5x = 15$
 $x = 3$ ✓

(d)(i) $\frac{(2x-1)^4}{2(4)} + c = \frac{1}{8}(2x-1)^4 + c$ ✓

(ii) $\left[-e^{-x}\right]_0^{\ln 2} = -e^{-\ln 2} - -e^{-0}$
 $= -\frac{1}{2} + 1$
 $= \frac{1}{2}$ ✓

Question 13

$$(a)(i) m_{AC} = \frac{4-0}{7-1} \\ = \frac{2}{3} \checkmark$$

$$(ii) \tan \theta = \frac{2}{3} \\ \therefore \theta = 34^\circ \checkmark$$

$$(iii) y-0 = \frac{2}{3}(x-1) \\ 2x-3y-2=0 \checkmark$$

$$(iv) D = \left(\frac{1+7}{2}, \frac{0+4}{2} \right) \\ = (4, 2) \checkmark$$

$$(v) m_{BD} = \frac{8-2}{0-4} \\ = -\frac{3}{2} \checkmark$$

$$m_{AC} \times m_{BD} = -1$$

$$(vi) AC = \sqrt{(7-1)^2 + (4-0)^2} \\ = 2\sqrt{13} \\ BD = \sqrt{(8-2)^2 + (0-4)^2} \\ = 2\sqrt{13} \checkmark$$

$$A = \frac{1}{2} \times 2\sqrt{13} \times 2\sqrt{13} \\ = 26u^2 \checkmark$$

$$(vii) E = (8, -4)$$

(b)(i) in ΔPBQ and ΔCQR

$$BQ = CR \text{ (given)}$$

$$\angle PBQ = \angle QCR = 90^\circ \text{ (square)}$$

$$BF = QC \text{ (length AB - PA = BC - BQ } \because \text{ square, PA = BQ)}$$

$$\therefore \Delta PBQ \equiv \Delta CQR \text{ (SAS or RHS)} \checkmark$$

$$(iii) \text{ let } \angle BQP = \alpha$$

$$\angle BPQ = 90 - \alpha \text{ (L sum } \Delta)$$

$$\angle CQR = 90 - \alpha \text{ (corrsp. } \angle \text{ s in cong. } \Delta \text{ s)} \checkmark$$

$$\angle BQP + \angle PQR + \angle CQR = 180^\circ \\ (\angle \text{ s on st. line}) \checkmark$$

$$\therefore \alpha + \angle PQR + 90 - \alpha = 180^\circ$$

$$\angle PQR = 90^\circ$$

$$(c) y = 2 \sec x \\ y^2 = 4 \sec^2 x$$

$$V = \pi \int y^2 dx \\ = \pi \int_0^{\frac{\pi}{3}} 4 \sec^2 x dx \checkmark$$

$$= 4\pi [\tan x]_0^{\frac{\pi}{3}} \checkmark$$

$$= 4\pi \left(\tan \frac{\pi}{3} - \tan 0 \right)$$

$$= 4\sqrt{3}\pi u^3 \checkmark$$

Question 14

$$(a)(i) -\frac{b}{a} = \frac{3}{2} \checkmark$$

$$(ii) \frac{c}{a} = -2 \checkmark$$

$$(iii) \frac{1}{\alpha} \times \frac{\beta}{\beta} + \frac{1}{\beta} \times \frac{\alpha}{\alpha} = \frac{\beta + \alpha}{\alpha\beta} = -\frac{3}{4} \checkmark$$

$$(iv) \alpha^2 + \beta^2 = (\alpha + \beta)^2 - 2\alpha\beta = \frac{25}{4} \checkmark$$

$$(b) \text{ LHS} = \sec x (\sec x + \tan x)$$

$$= \frac{1}{\cos x} \left(\frac{1}{\cos x} + \frac{\sin x}{\cos x} \right)$$

$$= \frac{1 + \sin x}{\cos^2 x} \checkmark$$

$$= \frac{1 + \sin x}{(1 + \sin x)(1 - \sin x)} \checkmark$$

$$= \frac{1}{1 - \sin x}$$

$$= \text{RHS.}$$

$$(c)(i) \frac{-\sin x}{\cos x} \checkmark = -\tan x$$

$$(ii) A = \int_0^{\frac{\pi}{4}} \tan x - x dx \checkmark \\ = \left[-\ln(\cos x) - \frac{x^2}{2} \right]_0^{\frac{\pi}{4}} \checkmark$$

$$= \left(\ln \frac{1}{\sqrt{2}} - \frac{\pi^2}{32} \right) - (-\ln 1 - 0) = \frac{1}{2} \ln 2 - \frac{\pi^2}{32} \checkmark \text{ or } \ln \sqrt{2} - \frac{\pi^2}{32}$$

$$\int_1^5 \log_e x dx = \frac{3-1}{6} (0 + 4 \log_e 3 + \log_e 5) \checkmark$$

$$= \frac{2}{3} (6.00) = 4.00 \checkmark$$

(d)(i) $m_{OP} = \frac{y}{x}$, $m_{PA} = \frac{y}{x-2}$

$$m_{OP} \times m_{PA} = -1$$

$$\therefore \frac{y}{x} \times \frac{y}{x-2} = -1 \checkmark$$

$$y^2 = -x^2 + 2x \checkmark$$

(ii) $x^2 - 2x + 1 + y^2 = 1 \checkmark$

$$(x+1)^2 + y^2 = 1$$

centre $(-1, 0)$ radius $= 1$ $\checkmark$

Question 15

(a) $5\pi = 6\theta$

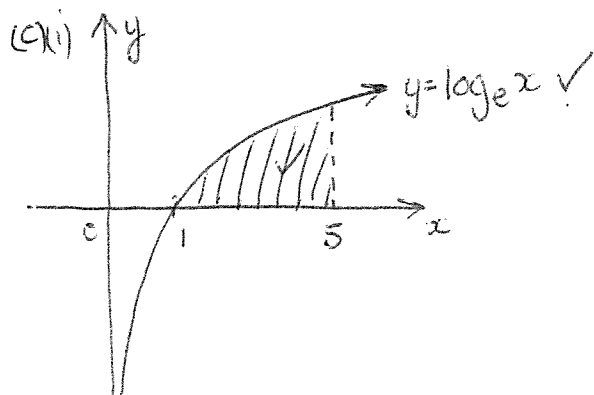
$$\theta = \frac{5\pi}{6} \checkmark$$

$$A_{AOB} = \frac{1}{2} (6)^2 \frac{5\pi}{6}$$

$$= 15\pi \text{ cm}^2 \checkmark$$

(b)(i) geometric $\checkmark$

(ii) $\left(4 + 3 + 2\frac{1}{4} + \dots \right)$
 $+ \left(3 + 2\frac{1}{4} + \dots \right) \checkmark$
 $S_{\infty} = 4 + 2 \left(\frac{3}{1 - \frac{3}{4}} \right)$
 $= 4 + 24$
 $= 28 \text{ m} \checkmark$



(ii)

x	1	3	5
y	0	1.10	1.61

(d) $-12 = 27a - 81 + b$ (turning pt.)

$$27a + b = 69$$

$$y' = 3ax^2 - 18x \text{ (minimum)} \checkmark$$

$$0 = 27a - 54$$

$$\therefore a = 2 \checkmark$$

$$b = 69 - 54$$

$$b = 15 \checkmark$$

Question 16

(a)(i) $A_1 = 400(1.005) = \$402 \checkmark$

(iii) $A_2 = (A_1 + 400)1.005 \checkmark$

$$A_2 = (400(1.005) + 400)1.005$$

$$A_2 = 400(1.005 + 1.005^2)$$

$$\vdots$$

$$A_n = 400(1.005 + 1.005^2 + \dots + 1.005^{12n})$$

(iii) sub in $n = 8$

$$A_8 = 400(1.005 + 1.005^2 + \dots + 1.005^{96}) \checkmark$$

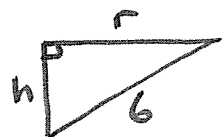
$$= \frac{400 \times 1.005(1.005^{96} - 1)}{0.005} \checkmark$$

$$= \$49377 \checkmark$$

(iv) $49377 = P(1.005)^{96} \checkmark$

$$P = \$30590 \checkmark$$

(b)(i) $V = \frac{1}{3} \pi r^2 h$



$$h^2 = 36 - r^2$$

$$h = \sqrt{36 - r^2} \checkmark$$

$$V = \frac{1}{3} \pi r^2 (\sqrt{36 - r^2}) \checkmark$$

$$\begin{aligned}
 \text{(ii)} \quad \frac{dV}{dr} &= (36-r^2)^{\frac{1}{2}} \times \frac{2}{3}\pi r \checkmark \\
 &\quad + \frac{1}{3}\pi r^2 \times \frac{1}{2}(36-r^2)^{-\frac{1}{2}} \times -2r \checkmark \\
 &= \frac{2\pi r \sqrt{36-r^2}}{3} - \frac{\pi r^3}{3\sqrt{36-r^2}} \checkmark
 \end{aligned}$$

$$\text{(iii)} \quad \frac{dV}{dr} = 0, \quad \frac{2\pi r \sqrt{36-r^2}}{3} = \frac{\pi r^3}{3\sqrt{36-r^2}}$$

$$\begin{aligned}
 \therefore 2(36-r^2) &= r^2 \\
 3r^2 &= 72 \\
 r &= 2\sqrt{6} \checkmark
 \end{aligned}$$

test:

r	0	$2\sqrt{6}$	5
$\frac{dV}{dr}$	>0	$=0$	<0

 $\checkmark$

$$\therefore \text{maximum at } r = 2\sqrt{6}$$

$$V = \frac{1}{3}\pi \times (2\sqrt{6})^2 \times \sqrt{36-(2\sqrt{6})^2}$$

$$\underline{V = 8\pi\sqrt{12} \text{ or } 16\pi\sqrt{3} \text{ u}^3 \checkmark}$$